

2106.07890: equation evidence

Display equations

| Identifier             | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2106.07890_E00004      | 7    | ■ 0.255 | $\begin{cases} * \rightarrow * & \text{when } d \text{ contains } c, \text{ red, and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does not contain blue.} \end{cases}$ | $\begin{cases} * \rightarrow * & \text{when } d \text{ contains } c, \text{ red, and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does not contain blue.} \end{cases}$ | $\begin{cases} * \rightarrow * & \text{when } d \text{ contains } c, \text{ red, and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does not contain blue.} \end{cases}$ |
| 2106.07890_E00012      | 11   | ■ 0.403 | $\begin{aligned} & \mathcal{L}(\text{red,red firetruck}) = 0.02 \\ & \mathcal{L}(\text{red,red idea}) = 10^{-5} \\ & \mathcal{L}(\text{red, blue sky}) = 0 \end{aligned}$                                                                                                                                                                                                          | $\begin{aligned} \mathcal{L}(\text{red,red firetruck}) &= 0.02 \\ \mathcal{L}(\text{red,red idea}) &= 10^{-5} \\ \mathcal{L}(\text{red, blue sky}) &= 0 \end{aligned}$                                                                                                                                                                                                             | $\begin{aligned} \mathcal{L}(\text{red,red firetruck}) &= 0.02 \\ \mathcal{L}(\text{red,red idea}) &= 10^{-5} \\ \mathcal{L}(\text{red,blue sky}) &= 0. \end{aligned}$                                                                                                                                                                                                             |
| 2106.07890_E00030      | 18   | ■ 0.508 | $\begin{aligned} & \mathcal{L}(c, d) \lim_W F(c) = \min \left\{ \frac{\mathcal{L}(c, d) f(c)}{w_1}, \frac{\mathcal{L}(c, d) g(c)}{w_2}, \mathcal{L}(c, d) \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & = \lim_W F(d) \end{aligned}$                                   | $\begin{aligned} \mathcal{L}(c, d) \lim_W F(c) &= \min \left\{ \frac{\mathcal{L}(c, d) f(c)}{w_1}, \frac{\mathcal{L}(c, d) g(c)}{w_2}, \mathcal{L}(c, d) \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & = \lim_W F(d) \end{aligned}$                                    | $\begin{aligned} \mathcal{L}(c, d) \lim_W F(c) &= \min \left\{ \frac{\mathcal{L}(c, d) f(c)}{w_1}, \frac{\mathcal{L}(c, d) g(c)}{w_2}, \mathcal{L}(c, d) \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & = \lim_W F(d), \end{aligned}$                                   |
| 2106.07890_E00024 (16) | 17   | ■ 0.538 | $\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\operatorname{op}}}(W, \mathcal{E}(F, -))$                                                                                                                                                                                                                                                               | $\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\operatorname{op}}}(W, \mathcal{E}(F, -))$                                                                                                                                                                                                                                                               | $\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\operatorname{op}}}(W, \mathcal{E}(F, -))$                                                                                                                                                                                                                                                               |

| Identifier             | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Image                                                                                                                                                                                                                                                                                                                                                     |
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| 2106.07890_E00034      | 19   | 0.573 | $\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) &= \inf_{c \in \mathcal{L}} \{[hc, (f \times g)c]\} \\ &= \inf_{c \in \mathcal{L}} \{[hc, \min\{fc, gc\}]\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \min_{\{\inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, 1 \right\}, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{hc}, 1 \right\}, 1\}} \\ &= \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$ | $\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) &= \inf_{c \in \mathcal{L}} \{[hc, (f \times g)c]\} \\ &= \inf_{c \in \mathcal{L}} \{[hc, \min\{fc, gc\}]\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \min_{\{\inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, 1 \right\}, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{hc}, 1 \right\}, 1\}} \\ &= \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$ |                                                                                                                                                                                                                                                                                                                                                           |
| 2106.07890_E00044      | 21   | 0.657 | $\begin{aligned} \widehat{\mathcal{L}}([h^x, h^y](c)) &= \inf_{d \in \mathcal{L}} \{[(h^c \times h^x)(d), h^y(d)]\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\pi(d c) \times \pi(d x)}, 1 \right\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}. \end{aligned}$                                                                                                                                                  | $\begin{aligned} [h^x, h^y](c) &= \widehat{\mathcal{L}}(h^c \times h^x, h^y) \\ &= \inf_{d \in \mathcal{L}} \{[(h^c \times h^x)(d), h^y(d)]\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\pi(d c) \times \pi(d x)}, 1 \right\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}. \end{aligned}$                                                                                                                        | $\begin{aligned} [h^x, h^y](c) &= \widehat{\mathcal{L}}(h^c \times h^x, h^y) \\ &= \inf_{d \in \mathcal{L}} \{[(h^c \times h^x)(d), h^y(d)]\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\pi(d c) \times \pi(d x)}, 1 \right\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}. \end{aligned}$ |
| 2106.07890_E00003 (2)  | 7    | 0.667 | $\left\{ (h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d) \right\}_{d \in \text{ob}(\mathcal{C})}$                                                                                                                                                                                                                                                                                                                                                                  | $\left\{ (h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d) \right\}_{d \in \text{ob}(\mathcal{C})}$                                                                                                                                                                                                                                                                                                                                                                  | $\left\{ (h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d) \right\}_{d \in \text{ob}(\mathcal{C})}$                                                                                                                                                                                                                                           |
| 2106.07890_E00020 (12) | 14   | 0.776 | $c \mapsto h^x(c) := \begin{cases} \pi(c x) & \text{if } x \leq c \\ 0 & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                                                                                                                                           | $c \mapsto h^x(c) := \begin{cases} \pi(c x) & \text{if } x \leq c \\ 0 & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                                                                                                                                           | $c \mapsto h^x(c) := \begin{cases} \pi(c x) & \text{if } x \leq c \\ 0 & \text{otherwise.} \end{cases}$                                                                                                                                                                                                                                                   |
| 2106.07890_E00032 (19) | 18   | 0.783 | $\left( (w_1, h^x) \times (w_2, h^y) \right)(c) = \begin{cases} \min \left\{ \frac{\pi(c x)}{w_1}, \frac{\pi(c y)}{w_2}, 1 \right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                             | $\left( (w_1, h^x) \times (w_2, h^y) \right)(c) = \begin{cases} \min \left\{ \frac{\pi(c x)}{w_1}, \frac{\pi(c y)}{w_2}, 1 \right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                             | $\left( (w_1, h^x) \times (w_2, h^y) \right)(c) = \begin{cases} \min \left\{ \frac{\pi(c x)}{w_1}, \frac{\pi(c y)}{w_2}, 1 \right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise.} \end{cases}$                                                                                                                                     |
| 2106.07890_E00005      | 7    | 0.797 | $[h^x, h^y](c) = \begin{cases} * & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                                                                               | $[h^x, h^y](c) = \begin{cases} * & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                                                                               | $[h^x, h^y](c) = \begin{cases} * & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise.} \end{cases}$                                                                                                                                                                                       |

| Identifier            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Rendered                                                                                                                                                                                                                                                                                                                                                                                    | Image                                                                                                                                                                                                                                                                                                                                                                                        |
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| 2106.07890_E00008 (6) | 10   | ■ 0.890 | $[a, b] := \begin{cases} b/a & \text{if } b < a \\ \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                                                                                                                                                                                                     | $[a, b] := \begin{cases} b/a & \text{if } b < a \\ 1 & \text{otherwise} \end{cases}.$                                                                                                                                                                                                                                                                                                       | $[a, b] := \begin{cases} b/a & \text{if } b < a, \\ 1 & \text{otherwise.} \end{cases}$                                                                                                                                                                                                                                                                                                       |
| 2106.07890_E00040     | 20   | ■ 0.955 | $\begin{aligned} & \operatorname{colim}_W F(c) = \max \left\{ \mathcal{L}(c, d) w_1 f(c), \mathcal{L}(c, d) w_2 g(c) \right\} \\ & \leq \max \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2} \right\} \\ & = \operatorname{colim}_W F(d), \end{aligned}$                                                                                                                                                                                                                                                                  | $\mathcal{L}(c, d) \operatorname{colim}_W F(c) = \max \{ \mathcal{L}(c, d) w_1 f(c), \mathcal{L}(c, d) w_2 g(c) \}$ $\leq \max \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2} \right\}$ $= \operatorname{colim}_W F(d)$                                                                                                                                                                         | $\mathcal{L}(c, d) \operatorname{colim}_W F(c) = \max \{ \mathcal{L}(c, d) w_1 f(c), \mathcal{L}(c, d) w_2 g(c) \}$ $\leq \max \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2} \right\}$ $= \operatorname{colim}_W F(d),$                                                                                                                                                                         |
| 2106.07890_E00048     | 23   | ■ 0.967 | $\begin{aligned} & \left( \left( w_1, f \right) \times \left( w_2, g \right) \right) (c) = -\ln \left( \left( \left( w_1', f' \right) \times \left( w_2', g' \right) \right) (c) \right) \\ & = -\ln \left( \min \left\{ \frac{f'c}{w_1'}, \frac{g'c}{w_2'}, 1 \right\} \right) \\ & = \max \left\{ -\ln \left( \frac{f'c}{w_1'} \right), -\ln \left( \frac{g'c}{w_2'} \right), 0 \right\} \\ & = \max \{ -\ln(f'c) + \ln(w_1'), -\ln(g'c) + \ln(w_2'), 0 \} \\ & = \max \{ fc - w_1, gc - w_2, 0 \}. \end{aligned}$ | $\begin{aligned} ((w_1, f) \times (w_2, g))(c) &= -\ln(((w_1', f') \times (w_2', g'))(c)) \\ &= -\ln\left(\min\left\{\frac{f'c}{w_1'}, \frac{g'c}{w_2'}, 1\right\}\right) \\ &= \max\left\{-\ln\left(\frac{f'c}{w_1'}\right), -\ln\left(\frac{g'c}{w_2'}\right), 0\right\} \\ &= \max\{-\ln(f'c) + \ln(w_1'), -\ln(g'c) + \ln(w_2'), 0\} \\ &= \max\{fc - w_1, gc - w_2, 0\} \end{aligned}$ | $\begin{aligned} ((w_1, f) \times (w_2, g))(c) &= -\ln(((w_1', f') \times (w_2', g'))(c)) \\ &= -\ln\left(\min\left\{\frac{f'c}{w_1'}, \frac{g'c}{w_2'}, 1\right\}\right) \\ &= \max\left\{-\ln\left(\frac{f'c}{w_1'}\right), -\ln\left(\frac{g'c}{w_2'}\right), 0\right\} \\ &= \max\{-\ln(f'c) + \ln(w_1'), -\ln(g'c) + \ln(w_2'), 0\} \\ &= \max\{fc - w_1, gc - w_2, 0\}. \end{aligned}$ |
| 2106.07890_E00029     | 17   | ■ 0.990 | $\begin{aligned} & \min \left\{ \left[ w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{Zc}, 1 \right\} \right], \left[ w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{Zc}, 1 \right\} \right] \right\} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}. \end{aligned}$                                                                                                                                                                                                     | $[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) = \min \left\{ \left[ w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{Zc}, 1 \right\} \right], \left[ w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{Zc}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}.$                                                        | $[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) = \min \left\{ \left[ w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{Zc}, 1 \right\} \right], \left[ w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{Zc}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}.$                                                         |

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| 2106.07890_E00037      | 19   | ■ 0.996 | $\begin{aligned} & \widehat{\operatorname{colim}_W F i, Z} = \inf_{c \in \mathcal{L}} \{ \max \{ w_1 f c, w_2 g c \}, Z c \} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{\max \{ w_1 f c, w_2 g c \}}, 1 \right\} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c} \right\}. \end{aligned}$                                                    | $\widehat{\mathcal{L}}(\operatorname{colim}_W F i, Z) = \inf_{c \in \mathcal{L}} \{ [\max \{ w_1 f c, w_2 g c \}, Z c] \}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{\max \{ w_1 f c, w_2 g c \}}, 1 \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c} \right\}.$                                  | $\widehat{\mathcal{L}}(\operatorname{colim}_W F i, Z) = \inf_{c \in \mathcal{L}} \{ [\max \{ w_1 f c, w_2 g c \}, Z c] \}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{\max \{ w_1 f c, w_2 g c \}}, 1 \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c} \right\}.$                                  |
| 2106.07890_E00039      | 20   | ■ 0.997 | $\begin{aligned} & \{ [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) \} = \min \left\{ \left[ w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{f c}, 1 \right\} \right], \left[ w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{g c}, 1 \right\} \right] \right\} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c}, 1 \right\}. \end{aligned}$ | $[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) = \min \left\{ \left[ w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{f c}, 1 \right\} \right], \left[ w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{g c}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c}, 1 \right\}.$ | $[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) = \min \left\{ \left[ w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{f c}, 1 \right\} \right], \left[ w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{g c}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c}, 1 \right\}.$ |
| 2106.07890_E00035      | 19   | ■ 0.999 | $\left( w_1, f \right) \sqcup \left( w_2, g \right) := \operatorname{colim}_W F.$                                                                                                                                                                                                                                                                                                     | $(w_1, f) \sqcup (w_2, g) := \operatorname{colim}_W F.$                                                                                                                                                                                                                                                                                      | $(w_1, f) \sqcup (w_2, g) := \operatorname{colim}_W F.$                                                                                                                                                                                                                                                                                      |
| 2106.07890_E00001      | 7    | ■ 1.000 | $\left[ h^{\text{red}}, h^{\text{blue}} \right] : \mathbf{C} \rightarrow \mathbf{Set}$                                                                                                                                                                                                                                                                                                | $\left[ h^{\text{red}}, h^{\text{blue}} \right] : \mathbf{C} \rightarrow \mathbf{Set}$                                                                                                                                                                                                                                                       | $[h^{\text{red}}, h^{\text{blue}}] : \mathbf{C} \rightarrow \mathbf{Set}$                                                                                                                                                                                                                                                                    |
| 2106.07890_E00010      | 10   | ■ 1.000 | $\begin{aligned} & a \times b := \min \{ a, b \} \\ & a \sqcup b := \max \{ a, b \}. \end{aligned}$                                                                                                                                                                                                                                                                                   | $a \times b := \min \{ a, b \}$ $a \sqcup b := \max \{ a, b \}.$                                                                                                                                                                                                                                                                             | $a \times b := \min \{ a, b \}$ $a \sqcup b := \max \{ a, b \}.$                                                                                                                                                                                                                                                                             |
| 2106.07890_E00002 (1)  | 7    | ■ 1.000 | $\left[ h^{\text{red}}, h^{\text{blue}} \right] (c) = \widehat{\mathcal{C}}(h^c, [h^{\text{red}}, h^{\text{blue}}]) = \widehat{\mathcal{C}}(h^c \times h^{\text{red}}, h^{\text{blue}}).$                                                                                                                                                                                             | $\left[ h^{\text{red}}, h^{\text{blue}} \right] (c) = \widehat{\mathcal{C}}(h^c, [h^{\text{red}}, h^{\text{blue}}]) = \widehat{\mathcal{C}}(h^c \times h^{\text{red}}, h^{\text{blue}}).$                                                                                                                                                    | $[h^{\text{red}}, h^{\text{blue}}](c) = \widehat{\mathcal{C}}(h^c, [h^{\text{red}}, h^{\text{blue}}]) = \widehat{\mathcal{C}}(h^c \times h^{\text{red}}, h^{\text{blue}}).$                                                                                                                                                                  |
| 2106.07890_E00006 (4)  | 9    | ■ 1.000 | $\begin{aligned} & 1 \leq \mathcal{C}(x, x) \\ & \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{aligned}$                                                                                                                                                                                                                                                    | $1 \leq \mathcal{C}(x, x)$ $\mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$                                                                                                                                                                                                                                              | $1 \leq \mathcal{C}(x, x)$ $\mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$                                                                                                                                                                                                                                              |
| 2106.07890_E00007 (5)  | 9    | ■ 1.000 | $x \otimes y \leq z \text{ if and only if } x \leq [y, z]$                                                                                                                                                                                                                                                                                                                            | $x \otimes y \leq z \text{ if and only if } x \leq [y, z]$                                                                                                                                                                                                                                                                                   | $x \otimes y \leq z \text{ if and only if } x \leq [y, z]$                                                                                                                                                                                                                                                                                   |
| 2106.07890_E00009 (7)  | 10   | ■ 1.000 | $a b \leq c \text{ if and only if } a \leq [b, c]$                                                                                                                                                                                                                                                                                                                                    | $ab \leq c \text{ if and only if } a \leq [b, c]$                                                                                                                                                                                                                                                                                            | $ab \leq c \text{ if and only if } a \leq [b, c]$                                                                                                                                                                                                                                                                                            |
| 2106.07890_E00011      | 11   | ■ 1.000 | $\mathcal{L}(x, y) := \pi(y \mid x),$                                                                                                                                                                                                                                                                                                                                                 | $\mathcal{L}(x, y) := \pi(y \mid x),$                                                                                                                                                                                                                                                                                                        | $\mathcal{L}(x, y) := \pi(y \mid x),$                                                                                                                                                                                                                                                                                                        |
| 2106.07890_E00013 (8)  | 11   | ■ 1.000 | $\pi(z \mid y) \pi(y \mid x) = \pi(z \mid x)$                                                                                                                                                                                                                                                                                                                                         | $\pi(z \mid y) \pi(y \mid x) = \pi(z \mid x)$                                                                                                                                                                                                                                                                                                | $\pi(z \mid y) \pi(y \mid x) = \pi(z \mid x)$                                                                                                                                                                                                                                                                                                |
| 2106.07890_E00014 (9)  | 12   | ■ 1.000 | $\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)$                                                                                                                                                                                                                                                                                                                                        | $\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)$                                                                                                                                                                                                                                                                                               | $\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)$                                                                                                                                                                                                                                                                                               |
| 2106.07890_E00015 (10) | 12   | ■ 1.000 | $\mathcal{D}^c(f, g) := \int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)$                                                                                                                                                                                                                                                                                                               | $\mathcal{D}^c(f, g) := \int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)$                                                                                                                                                                                                                                                                      | $\mathcal{D}^c(f, g) := \int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)$                                                                                                                                                                                                                                                                      |



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|------------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2106.07890_E00016 (11) | 13   | ■ 1.000 | $\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{[fc, gc]\}$                                                                                                                                      | $\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{[fc, gc]\}$                                                                                                                                      | $\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{[fc, gc]\}$                                                                                                                                      |
| 2106.07890_E00017      | 13   | ■ 1.000 | $h^x := \mathcal{C}(x, -)$                                                                                                                                                                                 | $h^x := \mathcal{C}(x, -)$                                                                                                                                                                                 | $h^x := \mathcal{C}(x, -)$                                                                                                                                                                                 |
| 2106.07890_E00018      | 13   | ■ 1.000 | $\mathcal{C}(c, d) \leq [h^x(c), h^x(d)]$                                                                                                                                                                  | $\mathcal{C}(c, d) \leq [h^x(c), h^x(d)]$                                                                                                                                                                  | $\mathcal{C}(c, d) \leq [h^x(c), h^x(d)]$                                                                                                                                                                  |
| 2106.07890_E00019      | 13   | ■ 1.000 | $\widehat{\mathcal{C}}(h^x, f) \leq [h^x(x), fx] = [1, fx] = fx$                                                                                                                                           | $\widehat{\mathcal{C}}(h^x, f) \leq [h^x(x), fx] = [1, fx] = fx$                                                                                                                                           | $\widehat{\mathcal{C}}(h^x, f) \leq [h^x(x), fx] = [1, fx] = fx$                                                                                                                                           |
| 2106.07890_E00021      | 15   | ■ 1.000 | $x \xrightarrow{a} y \quad h^y \xrightarrow{a} h^x$                                                                                                                                                        | $x \xrightarrow{a} y \quad h^y \xrightarrow{a} h^x$                                                                                                                                                        | $x \xrightarrow{a} y \quad h^y \xrightarrow{a} h^x$                                                                                                                                                        |
| 2106.07890_E00022 (13) | 16   | ■ 1.000 | $\mathrm{C}(-, \lim F) \cong \mathrm{Set}^J(*, \mathrm{C}(-, F))$                                                                                                                                          | $\mathrm{C}(-, \lim F) \cong \mathrm{Set}^J(*, \mathrm{C}(-, F))$                                                                                                                                          | $\mathrm{C}(-, \lim F) \cong \mathrm{Set}^J(*, \mathrm{C}(-, F))$                                                                                                                                          |
| 2106.07890_E00023 (15) | 16   | ■ 1.000 | $\mathcal{E}(-, \lim_W F) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$                                                                                                                           | $\mathcal{E}(-, \lim_W F) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$                                                                                                                           | $\mathcal{E}(-, \lim_W F) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$                                                                                                                           |
| 2106.07890_E00025      | 17   | ■ 1.000 | $(w_1, f) \times (w_2, g) := \lim_W F$                                                                                                                                                                     | $(w_1, f) \times (w_2, g) := \lim_W F$                                                                                                                                                                     | $(w_1, f) \times (w_2, g) := \lim_W F$                                                                                                                                                                     |
| 2106.07890_E00026 (17) | 17   | ■ 1.000 | $\widehat{\mathcal{L}}(Z, \lim_W F) \cong [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F))$                                                                                                            | $\widehat{\mathcal{L}}(Z, \lim_W F) \cong [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F))$                                                                                                            | $\widehat{\mathcal{L}}(Z, \lim_W F) \cong [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F))$                                                                                                            |
| 2106.07890_E00027      | 17   | ■ 1.000 | $\inf_{c \in \mathcal{L}} \left\{ \left[ Zc, \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}$ | $\inf_{c \in \mathcal{L}} \left\{ \left[ Zc, \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}$ | $\inf_{c \in \mathcal{L}} \left\{ \left[ Zc, \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}$ |
| 2106.07890_E00028      | 17   | ■ 1.000 | $\widehat{\mathcal{L}}(Z, Fi) = \inf_{c \in \mathcal{L}} \{[Zc, Fic]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Fic}{Zc}, 1 \right\}$                                                                      | $\widehat{\mathcal{L}}(Z, Fi) = \inf_{c \in \mathcal{L}} \{[Zc, Fic]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Fic}{Zc}, 1 \right\}$                                                                      | $\widehat{\mathcal{L}}(Z, Fi) = \inf_{c \in \mathcal{L}} \{[Zc, Fic]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Fic}{Zc}, 1 \right\}$                                                                      |
| 2106.07890_E00031 (18) | 18   | ■ 1.000 | $((w_1, h^x) \times (w_2, h^y))(c) = \min \left\{ \frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1 \right\}$                                                                                                      | $((w_1, h^x) \times (w_2, h^y))(c) = \min \left\{ \frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1 \right\}$                                                                                                      | $((w_1, h^x) \times (w_2, h^y))(c) = \min \left\{ \frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1 \right\}$                                                                                                      |
| 2106.07890_E00033      | 18   | ■ 1.000 | $\widehat{\mathcal{L}}(h, f \times g) = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g)$                                                                                                    | $\widehat{\mathcal{L}}(h, f \times g) = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g)$                                                                                                    | $\widehat{\mathcal{L}}(h, f \times g) = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g)$                                                                                                    |
| 2106.07890_E00036 (20) | 19   | ■ 1.000 | $\widehat{\mathcal{L}}(\mathrm{colim}_W F, Z) = [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z))$                                                                                                      | $\widehat{\mathcal{L}}(\mathrm{colim}_W F, Z) = [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z))$                                                                                                      | $\widehat{\mathcal{L}}(\mathrm{colim}_W F, Z) = [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z))$                                                                                                      |
| 2106.07890_E00038      | 19   | ■ 1.000 | $\widehat{\mathcal{L}}(Fi, Z) = \inf_{c \in \mathcal{L}} \{[Fic, Zc]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Zc}{Fic}, 1 \right\}$                                                                      | $\widehat{\mathcal{L}}(Fi, Z) = \inf_{c \in \mathcal{L}} \{[Fic, Zc]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Zc}{Fic}, 1 \right\}$                                                                      | $\widehat{\mathcal{L}}(Fi, Z) = \inf_{c \in \mathcal{L}} \{[Fic, Zc]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Zc}{Fic}, 1 \right\}$                                                                      |
| 2106.07890_E00041      | 20   | ■ 1.000 | $[f, g](c) := \widehat{\mathcal{L}}(h^c \times f, g)$                                                                                                                                                      | $[f, g](c) := \widehat{\mathcal{L}}(h^c \times f, g)$                                                                                                                                                      | $[f, g](c) := \widehat{\mathcal{L}}(h^c \times f, g)$                                                                                                                                                      |
| 2106.07890_E00042      | 21   | ■ 1.000 | $\widehat{\mathcal{L}}(h \times f, g) = \widehat{\mathcal{L}}(h, [f, g])$                                                                                                                                  | $\widehat{\mathcal{L}}(h \times f, g) = \widehat{\mathcal{L}}(h, [f, g])$                                                                                                                                  | $\widehat{\mathcal{L}}(h \times f, g) = \widehat{\mathcal{L}}(h, [f, g])$                                                                                                                                  |
| 2106.07890_E00043 (21) | 21   | ■ 1.000 | $[h^x, h^y](c) = \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}$                                                                                                 | $[h^x, h^y](c) = \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}$                                                                                                 | $[h^x, h^y](c) = \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}$                                                                                                 |

| Identifier        | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                                                        | Image                                                                                                                                                                                                                                                                                           |
|-------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2106.07890_E00045 | 23   | ■ 1.000 | $\widehat{\mathcal{M}}(f, g) = \sup_{x \in \mathcal{M}} d_{[0, \infty]}(fx, gx) = \sup_{x \in \mathcal{M}} \max\{gx - fx, 0\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                 |
| 2106.07890_E00046 | 23   | ■ 1.000 | $\begin{gathered} \left( \left( \left( w_1, f \right) \times \left( w_2, g \right) \right) \right) (c) = \max \left\{ f c - w_1, g c - w_2, 0 \right\} \\ \left( \left( w_1, f \right) \sqcup \left( w_2, g \right) \right) (c) = \min \left\{ f c + w_1, g c + w_2 \right\} \end{gathered}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} ((w_1, f) \times (w_2, g))(c) &= \max\{fc - w_1, gc - w_2, 0\} \\ ((w_1, f) \sqcup (w_2, g))(c) &= \min\{fc + w_1, gc + w_2\} \end{aligned}$                                                                                                                                   | $\begin{aligned} ((w_1, f) \times (w_2, g))(c) &= \max\{fc - w_1, gc - w_2, 0\} \\ ((w_1, f) \sqcup (w_2, g))(c) &= \min\{fc + w_1, gc + w_2\} \end{aligned}$                                                                                                                                   |
| 2106.07890_E00047 | 23   | ■ 1.000 | $\left( \left( \left( w_1^{\prime}, f^{\prime} \right) \times \left( w_2^{\prime}, g^{\prime} \right) \right) \right) (c) = \min \left\{ \frac{f^{\prime} c}{w_1^{\prime}}, \frac{g^{\prime} c}{w_2^{\prime}}, 1 \right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $((w_1', f') \times (w_2', g'))(c) = \min \left\{ \frac{f'c}{w_1'}, \frac{g'c}{w_2'}, 1 \right\}$                                                                                                                                                                                               | $((w_1', f') \times (w_2', g'))(c) = \min \left\{ \frac{f'c}{w_1'}, \frac{g'c}{w_2'}, 1 \right\}$                                                                                                                                                                                               |
| 2106.07890_E00049 | 24   | ■ 1.000 | $\begin{aligned} &\left( \left( \left( w_1, f \right) \times \left( w_2, g \right) \right) \right) (c) \text{ and } \left( \left( \left( w_1^{\prime}, f^{\prime} \right) \times \left( w_2^{\prime}, g^{\prime} \right) \right) \right) (c) \\ &\text{and } \left( \left( \left( w_1, f \right) \sqcup \left( w_2, g \right) \right) \right) (c) \text{ and } \left( \left( \left( w_1^{\prime}, f^{\prime} \right) \sqcup \left( w_2^{\prime}, g^{\prime} \right) \right) \right) (c) \\ &= \min \left\{ \left( \left( \left( w_1, f \right) \times \left( w_2, g \right) \right) \right) (c), \left( \left( \left( w_1^{\prime}, f^{\prime} \right) \times \left( w_2^{\prime}, g^{\prime} \right) \right) \right) (c), \right. \\ &\quad \left. \left( \left( \left( w_1, f \right) \sqcup \left( w_2, g \right) \right) \right) (c), \left( \left( \left( w_1^{\prime}, f^{\prime} \right) \sqcup \left( w_2^{\prime}, g^{\prime} \right) \right) \right) (c) \right\} \end{aligned}$ | $\begin{aligned} ((w_1, f) \sqcup (w_2, g))(c) &= -\ln(((w_1', f') \sqcup (w_2', g'))(c)) \\ &= -\ln(\max\{w_1' f' c, w_2' g' c\}) \\ &= \min\{-\ln(w_1' f' c), -\ln(w_2' g' c)\} \\ &= \min\{-\ln(f' c) - \ln(w_1'), -\ln(g' c) - \ln(w_2')\} \\ &= \min\{fc + w_1, gc + w_2\}. \end{aligned}$ | $\begin{aligned} ((w_1, f) \sqcup (w_2, g))(c) &= -\ln(((w_1', f') \sqcup (w_2', g'))(c)) \\ &= -\ln(\max\{w_1' f' c, w_2' g' c\}) \\ &= \min\{-\ln(w_1' f' c), -\ln(w_2' g' c)\} \\ &= \min\{-\ln(f' c) - \ln(w_1'), -\ln(g' c) - \ln(w_2')\} \\ &= \min\{fc + w_1, gc + w_2\}. \end{aligned}$ |
| 2106.07890_E00050 | 24   | ■ 1.000 | $s_1 \oplus s_2 = \min\{s_1, s_2\} \text{ and } s_1 \odot s_2 = s_1 + s_2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $s_1 \oplus s_2 = \min\{s_1, s_2\} \text{ and } s_1 \odot s_2 = s_1 + s_2$                                                                                                                                                                                                                      | $s_1 \oplus s_2 = \min\{s_1, s_2\} \text{ and } s_1 \odot s_2 = s_1 + s_2$                                                                                                                                                                                                                      |
| 2106.07890_E00051 | 25   | ■ 1.000 | $\left( \left( \left( w_1, f \right) \times \left( w_2, g \right) \right) \right) (c) = \min\{fc + w_1, gc + w_2\} = w_1 \odot fc \oplus w_2 \odot gc.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $((w_1, f) \sqcup (w_2, g))(c) = \min\{fc + w_1, gc + w_2\} = w_1 \odot fc \oplus w_2 \odot gc.$                                                                                                                                                                                                | $((w_1, f) \sqcup (w_2, g))(c) = \min\{fc + w_1, gc + w_2\} = w_1 \odot fc \oplus w_2 \odot gc.$                                                                                                                                                                                                |